COMPUTER PROGRAMMING

LAB NO: 10

LAB TITLE

More About Functions in C++

OBJECTIVE

The objective of this lab is to understand the concept of functions in C++. This lab explains function call by value, function call by reference, and default parameters. It also helps students learn how functions make programs easier, shorter, and more organized.

THEORY

A function is a separate block of code used to perform a specific task in a program. Functions reduce repetition in programming and improve readability of code. In C++, functions can receive values from the main program and can also return values after processing.

Example

```
#include <iostream>
using namespace std;
void message()
{
cout << "Welcome to Functions";
}
int main()
{
message();
return 0;
}
```

Function Call by Value

In call by value, a copy of the variable is sent to the function. Changes made inside the function do not affect the original variable in the main function.

Syntax

functionName(variable);

Function Call by Reference

In call by reference, the original variable is passed to the function using the & symbol. Any changes made inside the function also change the original value.

Syntax

void functionName(int &variable);

Default Parameters

Default parameters are values already assigned in the function declaration. If the user does not provide values during function calling, the default values are used automatically.

Syntax

int add(int a = 5, int b = 2);

SOFTWARE

DEV-C++ IDE

C++ Compiler

LAB TASKS

Function with Parameters ▪ Declare a function called "multiply" that takes two integer parameters:"num1"and"num2". ▪ Inside the function, calculate the product of the two numbers. ▪ Display the result within the function. ▪ Call the function from the main program, passing two numbers, to perform the multiplication.

```
#include <iostream>

using namespace std;

void multiply(int num1, int num2)// Ye function do numbers multiply kare ga

{

int product;
```

```
product = num1 * num2; // Dono numbers ka product calculate karna

cout << "Product = " << product << endl; // Result display karna

}

int main() {

int a = 5, b = 4;

multiply(a, b);// Function call karna

return 0;

}
```

Function with Return Value ▪ Declare a function called ”getSquare” that takes an integer parameter: ”number”. ▪ Inside the function, calculate the square of the number. ▪ Return the result. ▪ Call the function from the main program, passing a number, and display the square.

```
#include <iostream>

using namespace std;

int getSquare(int number)// Ye function square return kare ga

{

return number * number;// Number ka square return karna

}

int main() {

int num = 6;

int square = getSquare(num);// Function ko call karna

cout << "Square = " << square << endl;    // Result display karna

return 0;

}
```

CONCLUSION

In this lab, we learned different types of functions in C++. We understood function call by value, function call by reference, and default parameters. We also learned how to create functions with parameters and return values.